

WOOMERA, Australia

WMO: 946590

Lat: 31.1558S

Lon: 136.8053E

Elev: 169

StdP: 99.32

Time Zone: 9.50 (AUA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 4.2	(c) 5.2	(d) -15.9	(e) 1.0	(f) 29.3	(g) -12.0	(h) 1.4	(i) 25.8	(j) 12.4	(k) 16.1	(l) 11.6	(m) 16.3	(n) 3.3	(o) 110	(p) 0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 14.1	(c) 41.2	(d) 18.8	(e) 39.2	(f) 18.1	(g) 37.5	(h) 17.9	(i) 22.6	(j) 30.7	(k) 21.4	(l) 31.1	(m) 20.3	(n) 31.2	(o) 5.8	(p) 350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	15.8	24.8	18.7	13.8	24.5	16.8	12.2	24.1	67.4	30.8	62.9	31.2	58.9	30.8	35.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.0	10.7	9.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			2.1	43.9	0.8	1.5	1.5	45.0	1.1	45.8	0.6	46.6	0.1	47.7
			0.3	26.2	1.0	4.2	-0.4	29.3	-1.0	31.8	-1.6	34.2	-2.3	37.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	27.7	26.9	23.8	19.8	15.4	12.2	11.8	13.2	16.7	19.9	23.1	25.5
	DBStd	6.63	4.35	4.29	3.95	3.40	2.89	2.13	2.17	2.78	3.60	4.22	4.53	4.44
	HDD10.0	12	0	0	0	0	0	4	6	2	0	0	0	0
	HDD18.3	786	0	0	3	22	100	184	203	161	76	31	6	1
	CDD10.0	3528	549	474	428	293	168	71	61	102	200	306	394	482
	CDD18.3	1262	291	241	173	65	10	1	0	3	26	78	151	224
	CDH23.3	16685	4166	3297	2019	634	79	4	3	43	354	1021	1988	3076
	CDH26.7	9175	2569	1949	1004	227	13	1	0	7	120	471	1051	1764
Wind	WSAvg	5.1	5.8	5.5	5.1	4.5	4.2	4.1	4.4	4.8	5.5	5.7	5.7	5.7
Precipitation	PrecAvg	183	15	19	8	12	18	15	13	13	16	15	17	15
	PrecMax	493	93	121	45	69	119	75	55	71	86	82	131	86
	PrecMin	53	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	89	19	27	11	17	22	16	13	13	17	19	21	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	42.9	38.9	34.7	28.6	23.1	23.1	27.2	33.6	37.5	40.9	42.8
		MCWB	19.3	20.7	17.9	16.7	13.8	13.4	12.2	12.4	15.6	16.2	17.2	18.9
	2%	DB	41.3	40.3	36.8	31.3	25.6	20.1	20.9	24.1	30.1	34.2	37.9	39.9
		MCWB	19.0	19.0	17.7	15.4	13.7	12.1	11.2	11.5	13.4	15.2	17.1	17.7
	5%	DB	39.3	38.0	34.3	29.2	23.2	18.3	18.8	21.9	27.3	32.0	35.1	37.5
		MCWB	18.5	18.5	17.1	14.8	13.0	11.1	10.4	10.6	12.7	14.5	15.9	17.3
	10%	DB	37.1	36.1	32.1	27.0	21.3	17.0	17.1	19.8	24.9	29.1	32.7	35.1
		MCWB	18.1	18.1	16.5	14.3	12.3	10.9	9.8	10.1	11.9	13.7	15.4	16.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	24.5	22.2	19.7	16.5	15.8	14.6	14.8	17.7	19.5	21.0	22.8
		MCDB	31.5	28.3	28.4	23.2	21.3	19.2	18.3	21.4	23.2	29.4	28.8	29.7
	2%	WB	22.4	22.6	20.5	18.2	15.2	14.0	12.6	12.9	15.3	17.1	19.1	21.2
		MCDB	30.7	32.3	29.5	23.7	21.6	17.7	17.8	20.1	24.6	28.5	27.7	29.6
	5%	WB	21.1	21.2	19.0	16.7	14.2	12.7	11.5	11.8	14.0	15.9	18.1	19.6
		MCDB	32.2	32.0	29.5	25.6	20.7	16.4	16.7	19.0	23.9	27.1	29.4	31.1
	10%	WB	20.0	20.0	17.9	15.6	13.3	11.7	10.6	11.0	12.9	14.8	17.1	18.4
		MCDB	33.1	31.6	29.4	24.7	19.6	15.6	15.7	17.7	22.6	26.5	29.2	31.4
Mean Daily Temperature Range	5% DB	MDBR	14.1	13.3	12.6	11.7	10.6	9.6	10.5	11.6	12.9	13.6	13.6	14.0
		MCDBR	16.3	15.4	14.9	14.6	12.7	11.4	13.1	14.5	16.8	17.1	16.5	16.4
		MCWBR	5.5	4.8	5.0	5.6	5.6	5.7	6.4	6.3	6.8	6.6	5.7	5.5
	5% WB	MCDBR	12.7	11.9	12.0	11.2	10.3	9.0	10.9	12.0	13.7	14.4	13.1	13.2
		MCWBR	6.3	4.6	4.9	5.2	5.2	5.5	6.1	5.9	6.6	6.3	5.6	5.8
Clear-Sky Solar Irradiance	taub	0.354	0.348	0.330	0.322	0.305	0.292	0.296	0.311	0.341	0.347	0.355	0.344	
	taud	2.433	2.473	2.499	2.509	2.531	2.558	2.533	2.473	2.391	2.382	2.396	2.440	
	Ebn at Noon	986	972	956	912	879	870	881	910	929	962	978	998	
	Edh at Noon	123	114	104	93	81	74	79	94	114	123	126	122	
All-Sky Solar Radiation	RadAvg	7.89	7.18	6.12	4.69	3.54	3.02	3.30	4.26	5.65	6.74	7.46	7.95	
	RadStd	0.38	0.39	0.35	0.22	0.19	0.15	0.13	0.18	0.28	0.34	0.32	0.31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.45	N/A	N/A	+1.10	N/A	N/A	N/A	N/A	+158	+107
Regional (3 neighbors)	+0.44	N/A	N/A	+1.06	N/A	N/A	N/A	N/A	+158	+107

Nomenclature: See separate page